Journal on Multimodal User Interfaces
https://doi.org/10.1007/s12193-020-00339-7

ORIGINAL PAPER

A BCI video game using neurofeedback improves the attention of children with autism

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Received: 15 January 2020 / Accepted: 9 July 2020

Abstract
Major usability and technical challenges have created mistrust of the potential of brain computer interfaces used to control video games in challenging environments like healthcare. Despite several studies showing low cost commercial headsets can read the brainwave patterns of its users with great potential for long term adoption; there are limited studies showing its efficacy in concrete healthcare scenarios. In our past work, we developed FarmerKeeper, a BCI using users' attention to control a runner videogame to support neurofeedback therapies with great usability and user experience. In this paper, beyond usability, we describe the results of a 10-week deployment study with 26 children with severe autism using FarmerKeeper as a tool to support the neurofeedback therapies of children with autism. Pre- and post-assessment evaluation indicate all children with autism improve their attention, attentional control and sustained attention. Two children with autism no longer showed attention impairments in the post-assessment evaluation. We closed discussing directions for future work and the potential benefits of this new generation of BCI videogames in healthcare scenarios.

Keywords Autism · Attention · Brain–computer interface · Neurofeedback · Video game

1 Introduction

The capabilities of brain–computer interfaces (BCI) to easily control video games has shown their potential in making neurofeedback training increasingly popular and possible at home [1]. Neurofeedback [1] has the ability to recondition and retrain brain activity in order to enhance cognitive functions for healthy people [2–4] and people with certain neurodevelopmental conditions (e.g., Attention Deficit Disorder, ADHD, learning disabilities, stroke, epilepsy, cognitive dysfunction, depression, anxiety, and others) [5, 6]. However, the major usability and technical challenges existing BCI videogames currently faced have resulted in the mistrust from healthcare providers, poor treatment adherence and limited adoption in specialized neurologic clinics.

For more than 2 years using a consumer-grade BCI headset, we have been studying if a BCI videogame we developed, called FarmerKeeper [7], can be used as a tool to support the neurofeedback therapies in a clinic caring children with autism. Our previous work has shown great adoption and user experience of the use of the BCI video game we developed in support of children with autism; but, it has shown limited impact in improving the symptoms associated to autism. In this paper, in contrast, we investigate the efficacy of BCI videogames in improving the attention of children with autism. We chose to study this population as the needs of children with autism heavily differ from those of neurotypical increasing the complexity of the feasibility and applicability of BCI video games in a concrete scenario. Also, the use of BCI videogames with individuals has not been widely explored in the literature (see Sect. 2). This paper reports the first study showing results from the efficacy of BCI videogames in autism.

Our work has the following contributions: (1) empirical evidence showing FarmerKeeper is an effective treatment tool to improve the attention during the Neurofeedback training of children with autism, (2) empirical evidence showing BCI video games properly designed outperforms the gold standard, and (3) a case study showing BCI headsets and BCI video games are appropriate tools to measure users' attention.

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Published online: 31 July 2020

2 Related work

Recent work in BCI video games has been aimed at uncovering a more understandable feedback [8, 9], overlaying current commercial videogames (e.g., Mandryk et al. [8]) or developing scratch video games (e.g., Dancing Robot and Brain Chi [10]). All these games use a commercial headset to obtain the brain activity and calculate the user's attention to control the BCI video game. For example, Mandryk et al. [8] superimposes a texture overlay varying the intensity of the obfuscation of the screen according to users' attention. An evaluation of the effectiveness of the add-on with 16 children with Fetal Alcohol Spectrum Disorder (FASD) shows it could reduce the symptoms of ADHD [9]. Similarly, in MindLight children use a one-channel, dry-sensor EEG headset to control a light that glows from the avatar's head in the game. The more relaxed players become, the brighter the "mindlight" shines; the light is the only way that players can see in a dark haunted house. The use on MindLight with 136 neurotypical children (8–13 years old) show a significant reduction in child' anxiety. Lim et al. [11, 12] BCI video games designed for individuals with development impairments significant improve inattentive and hyperactive-impulsive symptoms on the ADHD rating scale. This body of work show modest efforts in improving some of the symptoms, specially attention, of neurotypical children or children with ADHD [9, 13, 14]; however, the widely held belief that consumer-grade headsets and BCI video games improve attention in the wider population lacks empirical support [15].

3 FarmerKeeper

FarmerKeeper [7] is a BCI video game developed to support neurofeedback training for children with autism. It uses the MindWave headset from Neurosky to read user's attention. The goal of the game is to help a little farmer to take lost farm animals back to their pens. The little farmer runs and pushes a truck to catch animals. The user's attention read with the headset controls the speed of the little farmer's truck, according to a threshold defined by the neurofeedback therapist. When the child has good attention (i.e., attention above the threshold), the little farmer will start to run and collects the lost animals appearing in the center of the road (Fig. 1, top-center). When the child loses attention, the little farmer will slow down. The volume of the background music will also gradually increase or decrease according to the player's attention. An old farmer appears to give the player short instructions when attention is very low for a period of time (Fig. 1, bottom-left) or if the signal is too noisy (Fig. 1, bottom-right). Once the newly recovered animals are back in their pens, the child can see them playing, grabbing a bite and drinking water. FarmerKeeper has five different scenarios, where the child could look for different farm animals like pigs, hens, cows, sheep, and horses. More details about the design and development of FarmerKeeper could be found in [7].

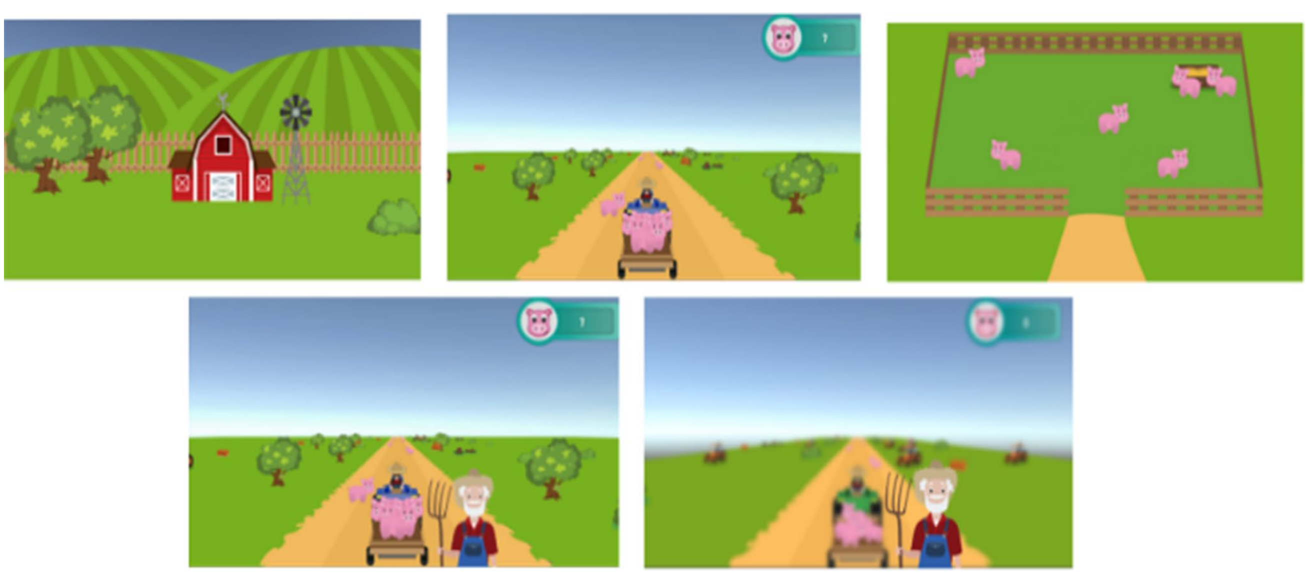

Fig. 1 FarmerKeeper's neurofeedback training section. Relax (top-left), activity (top-center) and break (top-right) phases. The activity phase during no-attention (bottom-left) and behavioral problems (bottom-right)

4 Evaluation methods

To evaluate the efficacy of FarmerKeeper in improving the attention of children with autism we conducted a between-subjects study at Pasitos, a school clinic specialized in the care of autism.

4.1 Participants

Twenty-six children with autism from Pasitos, a school clinic specialized in the care of autism, were voluntarily enrolled in this study. At the time of the study, children participants aged between four and 13 years old (avg. age = 8 years; SD = 3.05 years). Most of the participants were non-verbal, with attention and hypersensitivity problems. None of them followed a pharmacological treatment or had taken Neurofeedback training before this study. Two therapists trained in using neurofeedback conducted the sessions for all the participants. Also, ten teachers attended some sessions to observe the improvement and behaviors of the participants during the neurofeedback sessions, to be able to answer an interview at the end of the study. All the Neurofeedback therapists, all the teachers, and all the parents on behalf of the participants, gave their consent to participate in the study.

4.2 Setting up and installation

We equipped two therapy rooms (1.83 m × 3.65 m) at Pasitos, with a desktop computer used by the therapists to control the BCI systems and the EEG data being recorded during the sessions. We instrumented the desktop computers with a second monitor to display the feedback to the participant (i.e., FarmerKeeper or Cartoons). Also, we equipped both rooms with a video camera (front-side view) to monitor participants' interactions and behaviors. Participants used the MindWave headset from Neurosky as the BCI headset during the Neurofeedback sessions. We removed all the potential available stimuli inside the rooms (e.g., furniture, frames, and visual supports) (Fig. 2).

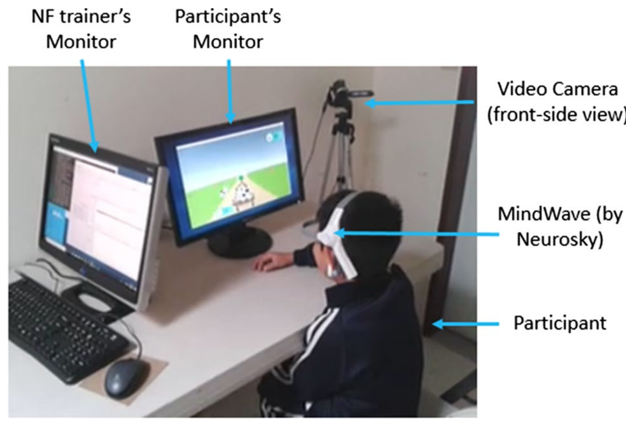

Fig. 2 A picture of a neurofeedback session showing the hardware installation in a therapy room at Pasitos for the study

4.3 Procedure

We randomly assigned the 26 participants into two groups of equal size (i.e., the experimental group and the control group). The experimental group of 13 participants used FarmerKeeper and the control group used Cartoons. We compared FarmerKeeper against Cartoons as they are widely used during Neurofeedback training and we are using them as our gold standard [1]. We used episodes from the "Untalkative Bunny" Cartoon as its animation is simple and uses visual and audible stimuli similar to those being used in FarmerKeeper. The selected episodes do not have dialogues to avoid confusion and increasing chances to make the cartoon more understandable for children participants.

Before the study, as most of the participants do not easily accept to wear accessories on their heads, we conducted a sensitization process to help them get used to wearing something similar to the MindWave headset. Participants wear on a daily basis a dummy-headset consisting of a pair of diadems assembled as the MindWave headset with a metallic button on the forehead and a clip on the ear lobe. During the study, participants completed 13 Neurofeedback sessions using Cartoons or FarmerKeeper, according to their assigned group. A Neurofeedback session lasted around 15 min, divided in three blocks of activity for around 4 min each, with breaks between activities lasting for 1–2 min each. Before and after the 13 neurofeedback sessions, children participants answered one assessment tests for attention (see Sect. 4.3.1). The assessment test lasted for around 13 min and were answered on a desktop PC using a two-button control we developed as the input interface instead of the keyboard to facilitate their interaction with the computer (see Sect. 4.3.1 for more details of the tests). To supplement this data, teachers answered another assessment test to evaluate the attention of participants, and were interviewed teachers on groups of two, to compare both conditions and gather their opinions and feedback. All the interviewed teachers attended some Neurofeedback sessions with participants.

4.3.1 Data collection

We collected assessment tests, self-reports, video, and EEG data.

We used the CRSD-ant [16] test and the ADHD-T [17] questionnaire as the assessment tests to evaluate attention pre and post-condition.

The CRSD-ant [16] is a 10-min version of the Attention Network Test [18] that measures three different functions of attention: alerting, orienting, and conflict monitoring, based on the attentional model proposed by Posner [19]. The CRSD-ant involves responding to the direction of a target object that is surrounded by flankers that point either in the same or opposite direction. Participants are instructed to respond as quickly as possible by pressing the left or right keys on the keyboard, according to the direction of the target object.

The ADHD-T [17] is a 36-item norm-referenced rating scale intended for use with children and youth ages 3–23 years. The ADHD-T consists of a single response form designed for use by teachers, parents, or other primary caregivers. Based on the DSM-IV [20] definition of ADHD, the measure consists of three subtests: Hyperactivity (13-item), Impulsivity (10-item), and Inattention (13-item). We only used the inattention sub-test, as we want to measure attention, and the design of FarmerKeeper is based on improving attention. The standard score of the inattention sub-test ranges from 1 to 15 points (with a mean = 10 and $\sigma = 3$). Where a low and high score indicates a low or high severity attention problems, respectively. This sub-test gives an awareness of the problems of attention in a general way, does not divide or specify subtypes of attention (i.e., selective attention, sustained attention, or others), and the questions of the sub-test correspond to activities of a daily basis.

We audio recorded and transcribed all the interviews (n = 6) for later analysis. All interviews lasted about 2:24:22 h. (avg. = 24:04 min, SD = 5:02 min). The semi-structured interview consisted of questions to gather information from participants, related to improvement in their attention or other behaviors. All neurofeedback sessions were video-recorded with a video camera (front-side view) to complement the self-reports from teacher and he Neurofeedback therapist. We recorded all the data received by the MindWave headset to analyze the attention of each participant during sessions.

4.3.2 Data analysis

We first removed data that included too much noise due to inappropriate body movements and gestural expressions. We also remove those participants that missed some sessions and not completed the 13 sessions. As a result, we had to remove data from three participants, two from the control group (cartoons) and one from the experimental group (FarmerKeeper).

From the EEG data recorded by the BCI headset, we extracted the attention data for each activity phase of each session per participant, per condition. Attention data is automatically calculated by the headset using its internal ThinkGear Chip, through its proprietary algorithm eSense.

To measure Attention, we computed the average of the percentage of time that the attention value was above or equal the predefined threshold (40-attention units) per session per participant, per condition. Using this data, we created a binary vector, per user, assigning 1 to the periods of time the attention value of such user was above 40 units, and 0 in the other case –meaning the user was distracted and his attention value was under 40.

To measure Attentional control, we used the attention binary vector to compute the number of changes between a no-attention value and an attention value during the attention phases per session and computed the average per participant per condition.

To measure Sustained Attention, we used the attention binary vector to compute the length of the attention blocks (i.e., seconds of consecutive attention values) per session and then the average length of an attention block per participant per condition.

We used the CRSD-ant software to automatically calculate the test scores per participant. We only took the results of the attention alert network, which is related to sustained attention. For the ADHD-T questionnaire, we calculated the raw and standard scores according to the process specified in the questionnaire's manual. We only took the scores of the inattention sub-test, as it measures the effect of the Neurofeedback sessions. For both tests, we computed the difference between the pre- and post-condition scores per participant.

For all of our data, and although our sample is small, we conducted a statistical analysis as a first step to test the significance of our results. As all of the collected data was non-parametric, according to the Shapiro–Wilk normality test [21], we used the Mann–Whitney U-test [22] with a confidence interval of $\alpha = 0.05$.

To analyze qualitative data from interviews, we used deductive analytical approaches to examine how observed behaviors and reported perceptions supported or contradicted our results from quantitative data. We additionally used inductive approaches to theme analysis from our data. To support our inductive analysis, we followed a qualitative approach including the use of techniques to derive grounded theory [23] and affinity diagramming [24] (e.g., open and axial coding [23]). Using these techniques, quotes and/or events obtained from interviews were grouped to uncover emergent themes related to improvements on attention, attentional control, and sustained attention by the Neurofeedback training. We used atlas.ti [25] for qualitative data analysis.

5 Results

Overall, our results show that participants that used FarmerKeeper slightly better improve their attention than participants who used Cartoons. FarmerKeeper participants showed a slightly greater improvement in attention, they were less dispersed, and achieved slightly longer attention blocks (i.e., they had better sustain attention), compared to the Cartoons participants. Also, FarmerKeeper participants showed slightly greater improvement on pre and post-condition psychometric attention.

5.1 Attention

Our EEG data shows, that on average participants in both conditions achieved a level of attention equal or greater than the defined threshold—almost 70% of the total time of the sessions (FarmerKeeper = 71.64%, Fig. 3 bottom left, and Cartoons = 68.73%, Fig. 3 top left). There is a slight difference of close to 3% ($p = 0.25$), which means that participants who used FarmerKeeper had a better performance than those who use Cartoons. Participants' attention value was above the threshold for 21 s more than those participants who used Cartoons (of a total of 720 s of the activity phases of the session). Also, FarmerKeeper participants (9 out of 12) had an equal or higher percentage of attention time than the average time of Cartoons participants (≥ 68.73%, Fig. 3 right); while most Cartoons participants (8 out of 11) focused equal or lower than the average percentage of time (≤ 71.64%, Fig. 3 right). These results suggest FarmerKeeper participants were slightly more focused during Neurofeedback sessions than Cartoons participants.

We attribute these results to the fact that the goal of the training is more clear in FarmerKeeper than in the Cartoons condition. Having a clearer goal, impacted in motivation as participants were more aware of how their performance will lead to achievements and rewards (e.g., better attention will increase changed to find more animals and coins). In contrast, Cartoons participants realized their performance did not change much, as no matter what, they will watch the Cartoon and apparently most of the time they did not care if they heard the sound or not.

All teachers (10 out of 10) agreed FarmerKeeper was more motivating for participants, and they perceived its goal to be clearer than the goal in the Cartoons condition. They corroborate our quantitative results explaining that this "goal awareness" indeed helped participants to be more focused.

These results suggest FarmerKeeper could be used as a more general tool, as it has a topic and an aim that could be more interesting to most participants.

5.2 Attentional control

On average, during the total time for all three activity phases, FarmerKeeper participants (Fig. 4, bottom) made fewer changes (8.6 fewer changes, $p = 0.04884$) between

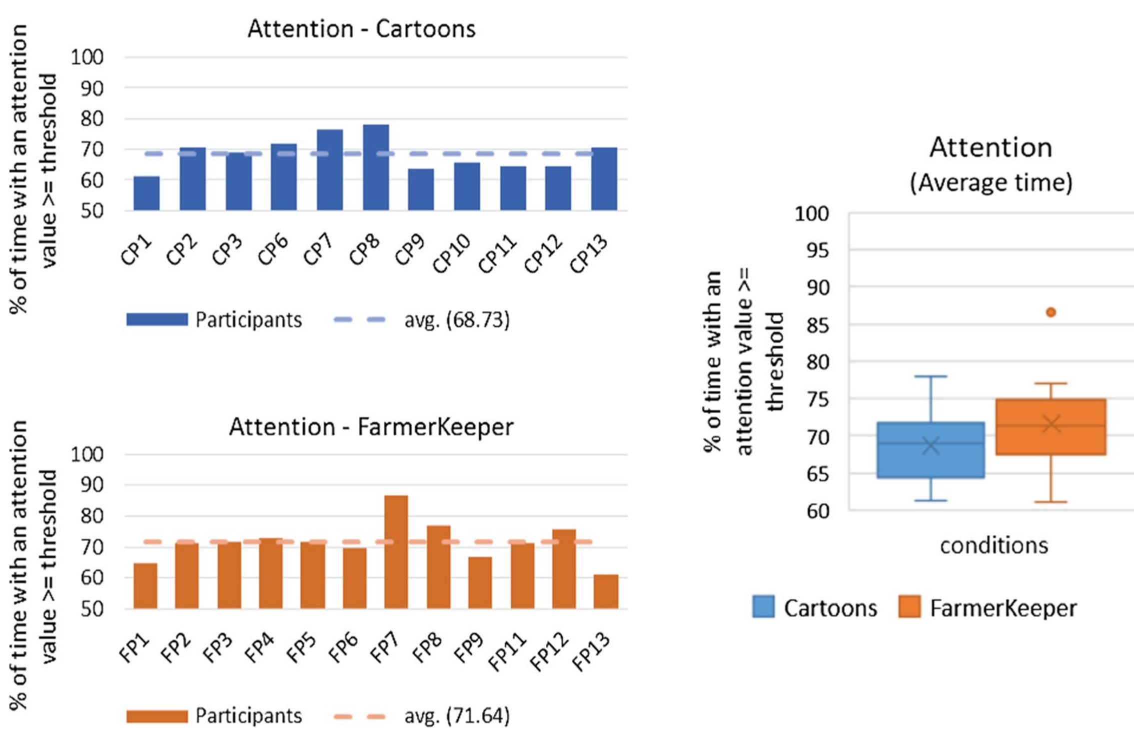

Fig. 3 (left) Average percentage of time during the activity phases with an attention value ≥ to the predefined threshold. Cartoons participants (top left) and FarmerKeeper participants (bottom left). (right) Comparison of the attention average time between Cartoons participants and FarmerKeeper participants

being distracted and focused, in contrast to Cartoons participants (Fig. 4, top). Also, most of FarmerKeeper participants (10 out of 12) made fewer changes than Cartoons participants; while most Cartoons participants (9 out of 11) made more changes than the average number of changes of FarmerKeeper participants (Fig. 4 right).

These results may suggest FarmerKeeper participants were less dispersed than the Cartoons participants. We attribute this behavior to the way the visual stimuli is being distributed in the FarmerKeeper main screen, avoiding to overload it and enabling participants to focus on only one part of the screen. By doing so, participants ended having a better performance without needing to juggle between too many distractions in different areas of the screen. In contrast, the visual stimuli in cartoons are scattered through the screen constantly competed for participants' attention making their attention to being more dispersed.

Teachers agreed FarmerKeeper has elements to keep them focused on the activity, like the animals and the position where the character's tractor moves. Also, they agreed that the simplicity of the arrangement of FarmerKeeper stimuli reduce dispersion.

This slightly better improvement in FarmerKeeper participants could be due to the fact that, on average, FarmerKeeper participants spent more time making an effort to achieve an attention value above the predefined threshold, in order to find more animals, while Cartoons participants easily lost the aim and they almost made no effort to achieve an attention value above the threshold. That is, the brain activity of the FarmerKeeper participant could self-regulated better during the sessions, which helped to improve the participant's attention in a better way, even outside the sessions.

All teachers (10 out of 10) agree FarmerKeeper can improve participant's attention better than Cartoons since when using FarmerKeeper participants make a greater effort than using Cartoons, which causes a greater improvement in attention.

5.3 Sustained attention

On average, during both conditions, participants consecutively maintained a level of attention for at least 10 s (Cartoons = 10.38 s, FarmerKeeper = 12.15 s; Fig. 5); meaning, their sustained attention is 10 s. In contrast, on average FarmerKeeper participants (Fig. 5, bottom) had 12 s of sustained attention ($p = 0.1164$; Fig. 5, top), 2 more sec. than Cartoons participants. Also, most FarmerKeeper participants (10 out of 12) achieve an equal or greater sustained attention than the average of the Cartoons participants. While most of the Cartoons participants (9 out of 11) achieve shorter

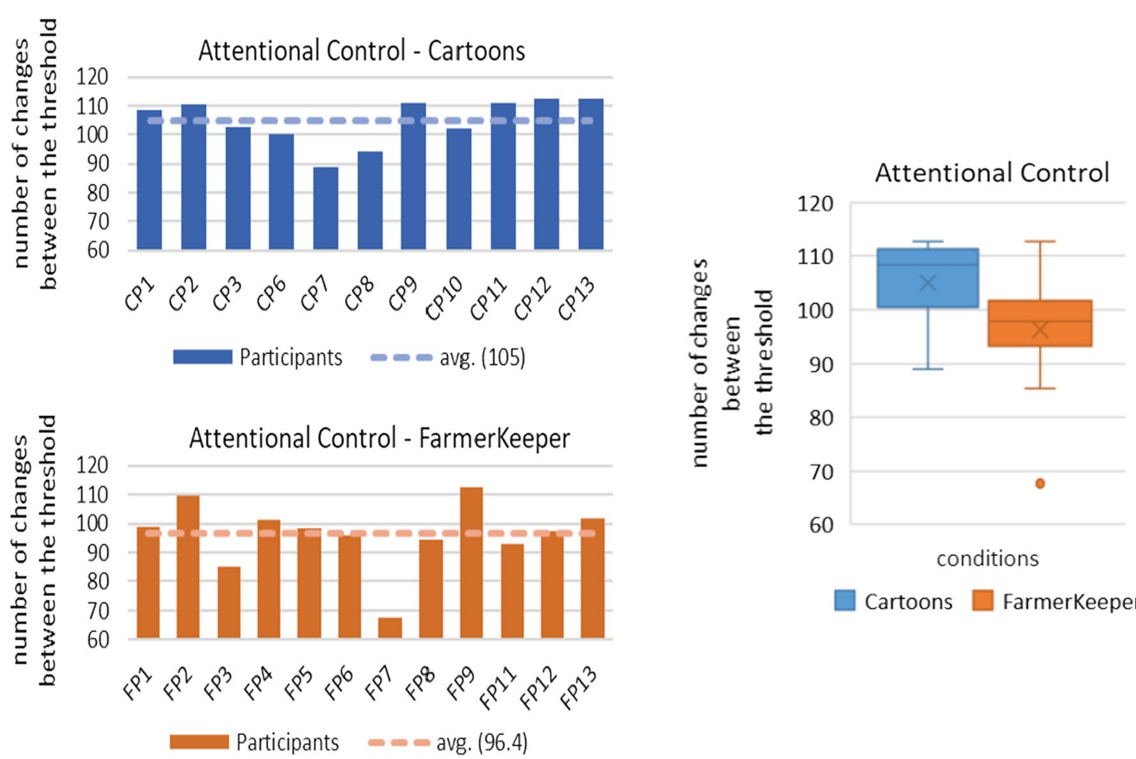

Fig. 4 Average number of changes between attention and no attention during the activity phases by participant. Cartoons participants (top) and FarmerKeeper participants (bottom). *A lower number of changes is better. (right) Comparison of the attentional control between Cartoons participants and FarmerKeeper participants. *A lower number of changes is better

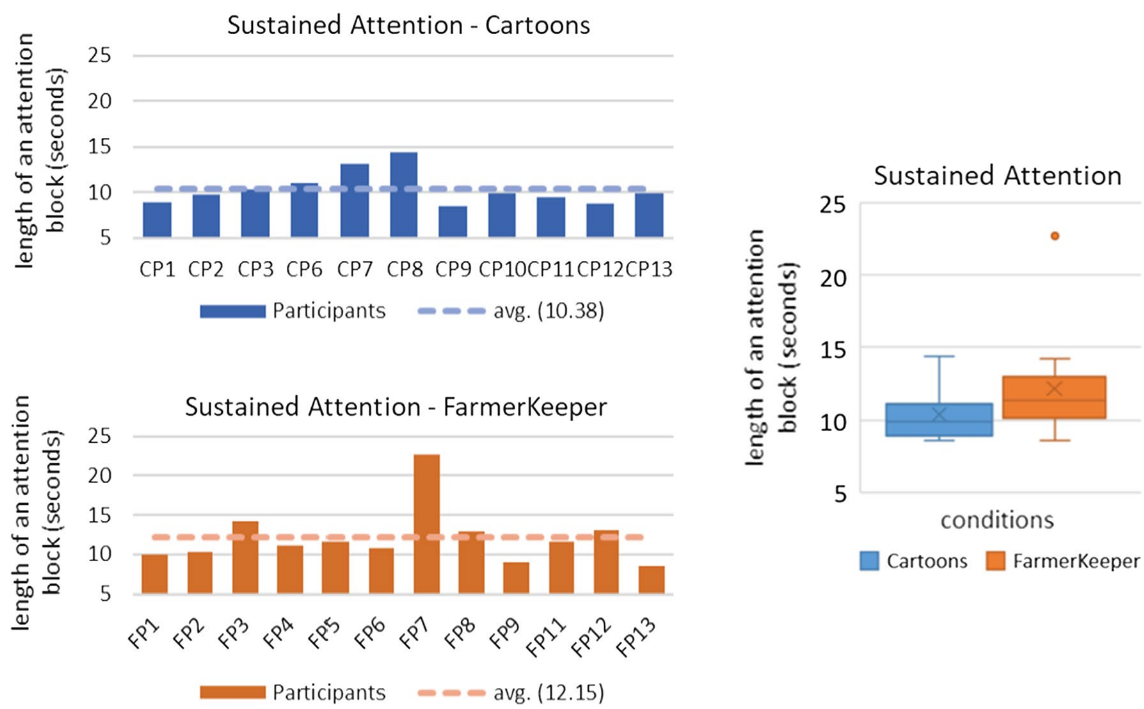

Fig. 5 Average sustained attention during the activity phases by participant. Cartoons participants (top) and FarmerKeeper participants (bottom). (right). Comparison of the sustained attention between Cartoons participants and FarmerKeeper participants

sustained attention than the average of the FarmerKeeper participants (Fig. 5 right). This could suggest FarmerKeeper participants achieved a slightly longer sustained attention than the Cartoon's participants did.

We attribute this to the capability of FarmerKeeper to maintain appropriate attention and attentional control, that will result in an increase sustained attention.

5.4 Pre- and post-condition psychometric attention tests

The results from the ADHD-T questionnaire corroborate our previous results, demonstrating that a greater number of FarmerKeeper participants (5 of 12) were less distracted than the Cartoons participants (2 of 11). At the same time, 2 out of 11 participants worsened their attention problems obtaining a higher score in the post-condition test when using Cartoons compared to only 1 out of 12 participants who used FarmerKeeper. The rest of the participants improved their attention. Comparing pre- and post-condition scores, we found out that the number of cartoons participants who improved or worsened their score were more dispersed than FarmerKeeper participants. This suggests that Cartoons may only work for participants with specific characteristics in contrast to FarmerKeeper that appears to fit the needs of most of the participants.

The results of the CRSD-ant test show both Cartoon (8 out 11) and FarmerKeeper participants (9 out of 12) improve their sustained attention. However, almost twice of FarmerKeeper participants (6 out of 12) achieve a score above the average score of neurotypicals [26]; in contrast, to Cartoons participants, where a smaller number (3 out of 11) presented a similar behavior. The pre- and post-condition scores of alert sub-test (i.e., sustained attention) show both FarmerKeeper and Cartoons participants increased their attention score, and the participants showing improvements were less dispersed. This could suggest Neurofeedback sessions had a good general improvement on participants' sustained attention, for both conditions.

Overall, these results suggest FarmerKeeper could improve and help to decrease participants' attention problems slightly better than Cartoons.

6 Limitations

Although, by improving the attention of children with autism we are not addressing core symptoms of autism; but a potential consequence of their condition, including their impairments in attention could serve as a vehicle to explore remaining open questions in this domain and deepen our understanding of the efficacy of BCI video games in a concrete scenario.

Some limitations in the study must be taken into account. One limitation is that the same threshold value was used for all participants in both conditions to avoid having more variables. As participants show progress, this challenge should be adjusted.

Another limitation is the use of the attention values given by the MindWave headset. A better algorithm tailored to the needs of children with autism, and to each patient's progress must be proposed.

Also, the sample is small, the results obtained are characteristic of the selected population, and we are not able to generalize them. The study must be replicated with a larger sample size to achieve a more confident level of generalization.

7 Conclusions and future work

FarmerKeeper participants show a slightly greater improvement in attention, they were less dispersed during the neurofeedback training, and they achieved a slightly longer sustained attention, than Cartoons participants. This could be attributed to a better understanding of the goal of the FarmerKeeper game, and its design, avoiding stimuli overload and enabling user to focus in one area of the screen At the same time, the topic used in FarmerKeeper is of interest to most participants, keeping them motivated during the neurofeedback training.

The results of the psychometric attention tests (i.e., CRSD-ant and ADHD-T) show FarmerKeeper participants slightly improve more their attention in similar activities (evaluated with the CRSD-ant test) and in the activities conducted in daily basis (evaluated with the ADHD-T test), than Cartoon's participants. We attribute this behavior, to the capability of FarmerKeeper to demand an appropriate effort to participants; in contrast to the Cartoons that were perceived not too demanding.

Overall, these results show FarmerKeeper could be a little bit more effective in helping to decrease attention problems than the gold standard used in Neurofeedback sessions.

As future work, we propose to explore the use of new algorithms to obtain the attention value and to measure the quality of the signal, to improve robustness. Also, we propose to explore the use of a dynamic threshold, which should be personalized to each participant, taking into account the tasks demands according to the participant's state and how it is affected by external factors (e.g., if the child slept well the previous day, if he is irritable, if he ate or not before the session, if he was a little indisposed).

Acknowledgements We thank all the participants enrolled in this study and the researchers and reviewers who provide helpful comments on previous versions of this document. We also thank CONACYT for the first author fellowship, and we thank the CONACYT Project #2209 of the third author for their financial support. (corresponding author: Monica Tentori).

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